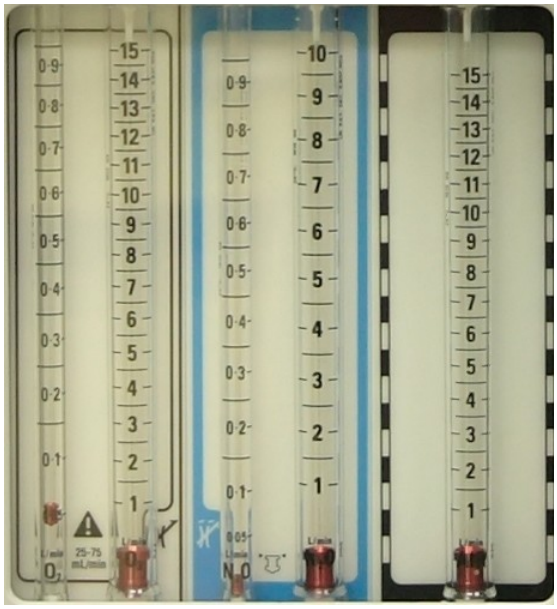
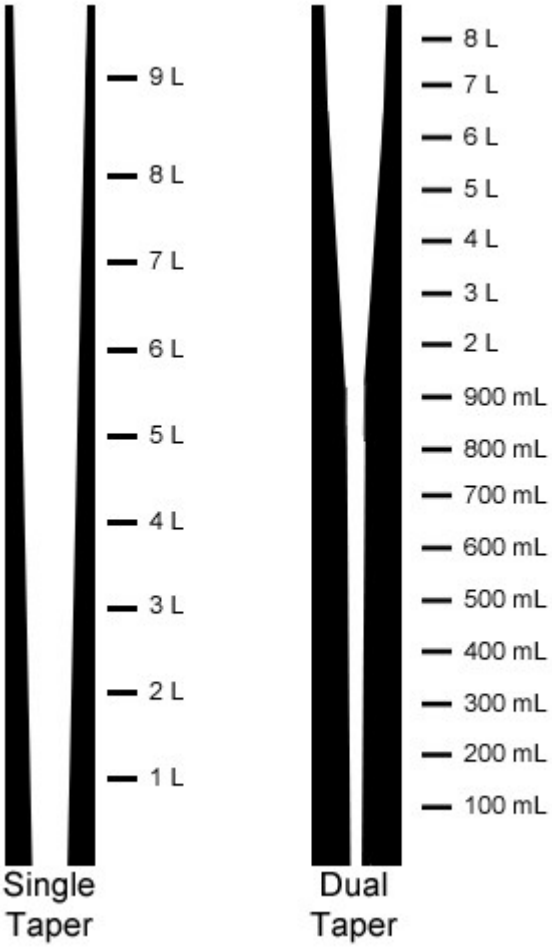
Flowmeters (also called rotameters, flow tubes, and flow indicators) display the amount of gas that is being delivered to the patient. Mechanical flowmeters like the ones pictured to the right function on the principle that flow past a resistance is proportion to the pressure. The flow meter actually measures the drop in pressure that occurs when the gas flow passes through a resistance.



Many newer anesthesia machines have electronic flowmeters that display a graphical representation of a real flowmeter. Often, there is also a mechanical flowmeter at the gas outlet to reassure anesthetists who do not trust electronic flowmeters.

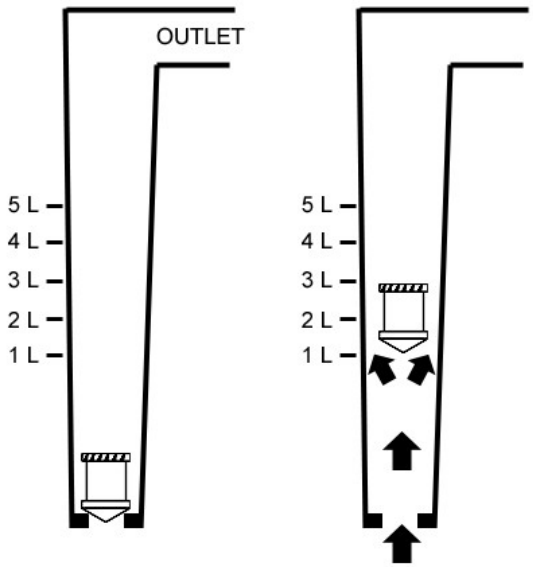


Most mechanical flowmeters are variable-orifice flowmeters (Thorpe tubes) that widen from bottom to top. When the control dial is turned, gas enters the flowmeter from the bottom and exits through an outlet at the top of the tube.

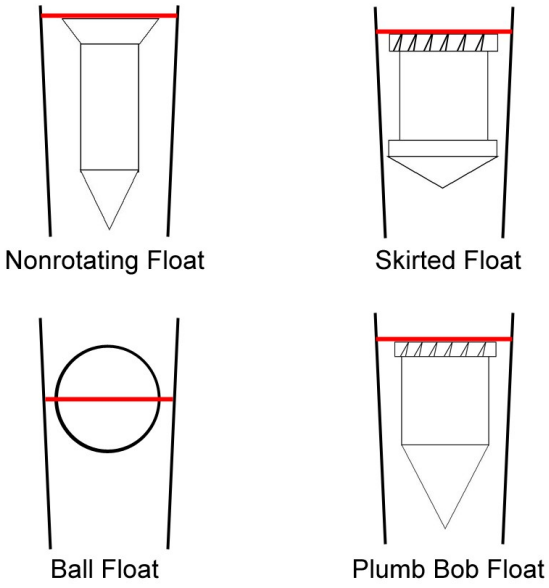
There are two types of variable-orifice tubes. Single-taper tubes have a steady and gradual increase in diameter from bottom to top and are usually used in anesthesia machines that have a separate tube for high and low flows. The dual-taper tube widens more quickly after it reaches 1 liter/min of gas flow and is capable of accurately measuring both low and high flows. The flowtubes empty into the common manifold that delivers the measured amount of gas into the low-pressure system of the anesthesia machine.

The flowmeter tubes are made of glass and are shielded in the machine behind a clear, plastic screen. A light is positioned on the anesthesia machine so that the flowmeters are visible in a dark operating room.

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In the flowtube on the left, no gas is entering the system and the indicator is resting at the bottom of the tube. When the gas flow valve is turned on and gas enters the tube, the indicator (also called a float or bobbin) floats freely in the tube.



Read the level at the red line

There are four types of indicators: nonrotating float, skirted float, ball float, and plumb bob float. The skirted float and plumb bob float are referred to as rotating indicators because they have vanes or grooves cut into the top portion that causes them to spin while they float in the tube. This allows the anesthetist to be certain that the float has not simply become stuck in the tube and is accurately measuring the gas flow. The worst case scenario would be a nonrotating or ball float stuck in the tube giving the impression that adequate oxygen flow is being delivered, when in fact the patient may be in jeopardy of hypoxia.

All of the indicators whould be read at their uppermost margin except for the ball float, which should be measured at the middle of the float.

Dorsch JA, Dorsch SE. Understanding Anesthesia Equipment. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins, 2008: 103-106.

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